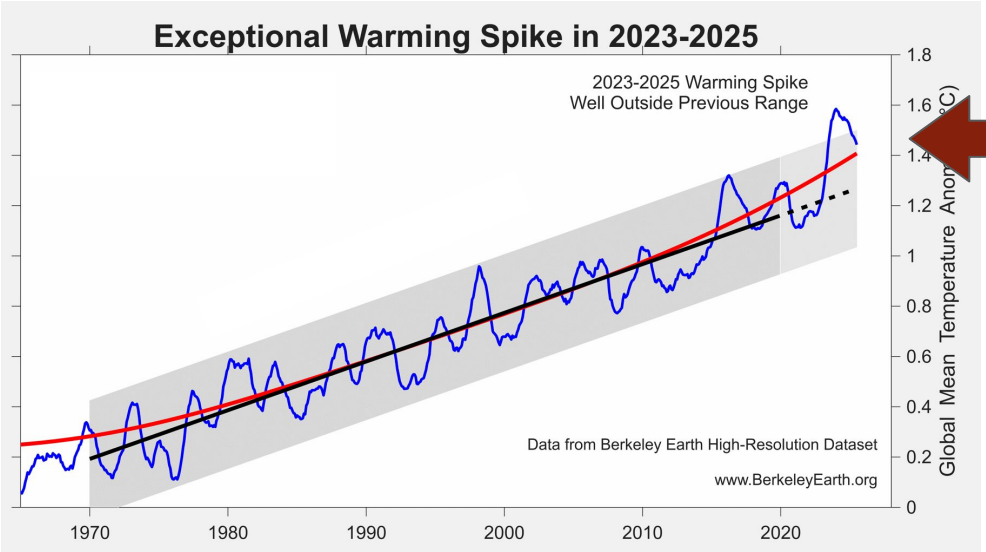
below the 1.5°C boundary

keeping the world cool

Recommendations to reduce Thailand's carbon emissions

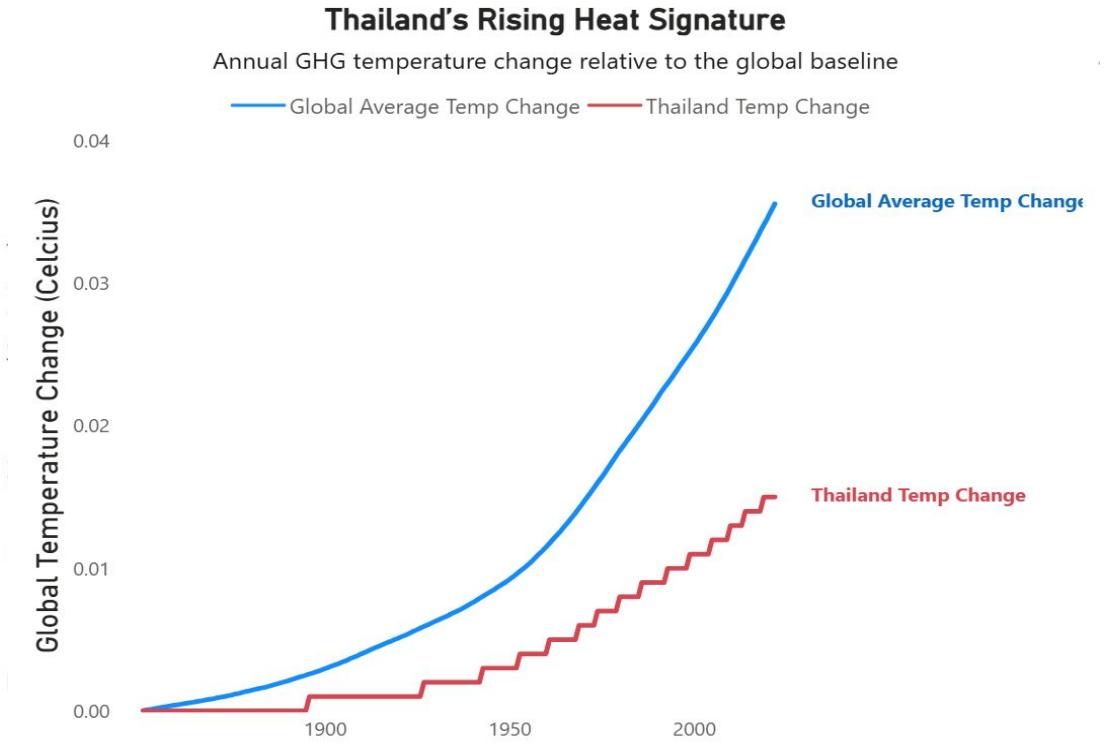
1.5° C

The earth is **no longer cool…**



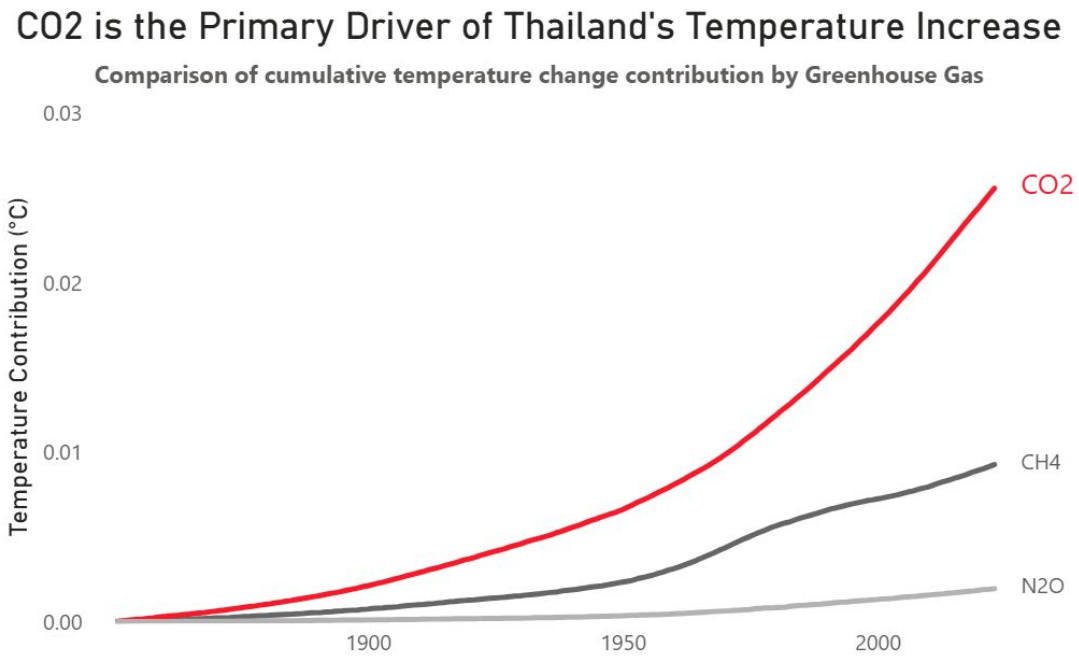
The global temperature continues to rise, and has spiked outside the previous range in the recent years…

...and Thailand is part of the problem.



GHG: greenhouse gases

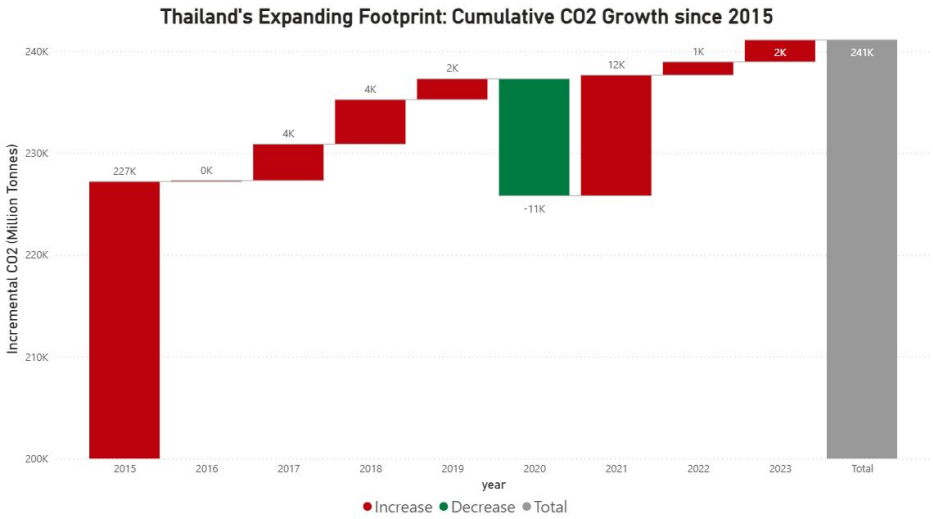
what is to blame?



CO2 = Carbon dioxide
CH4 = Methane
N2O = Nitrous oxide

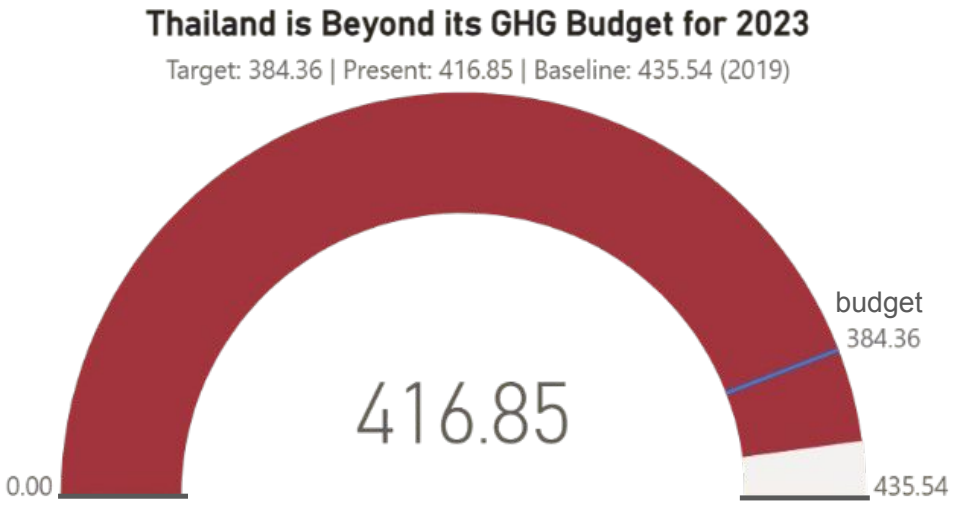
Carbon dioxide emissions primarily contribute to the increase in temperature

Our carbon emissions are increasing...



Carbon dioxide emissions continue to increase despite the Paris agreement in 2015.

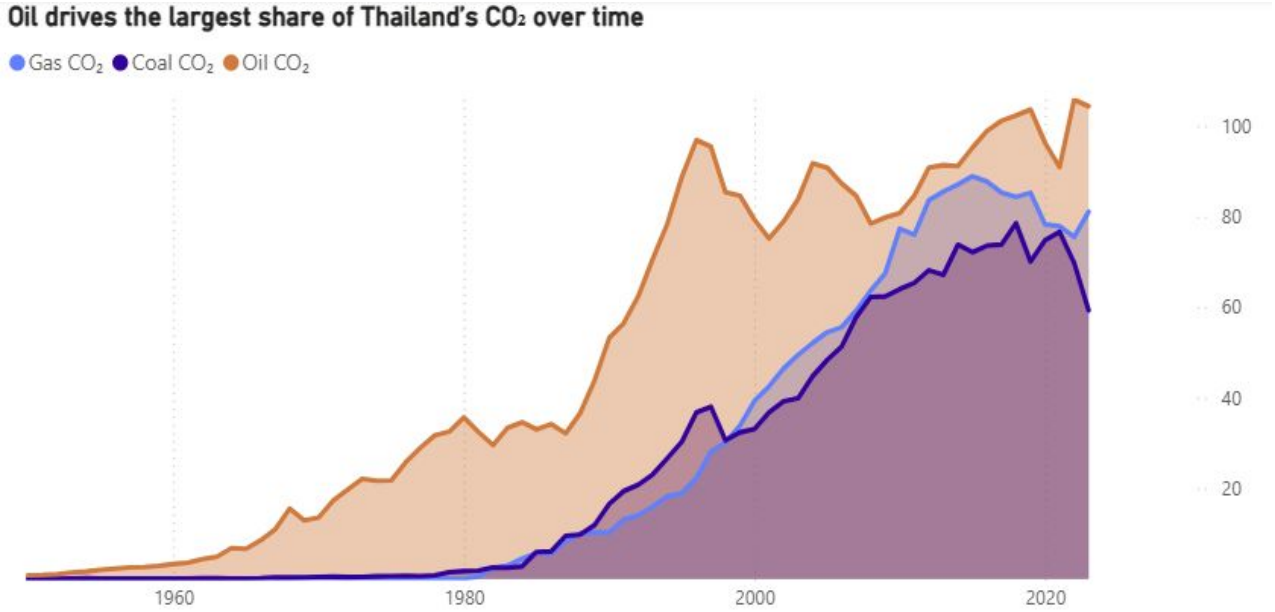
...but increased carbon emissions are something **we cannot afford.**



GHG budget = the maximum GHG emissions allowed to limit our temperature to 1.5°C

Emissions projections for 2030 and absolute differences in emissions are based on emissions of 52–56 GtCO2-eq yr–1 in 2019 as assumed in underlying model studies.

Oil and gas is fueling our spending.



There is a continued reliance on oil and gas despite initiatives to shift to clean energy

Sectors

35% Energy
32% Transport
27% Industrial
6% Residential

Saving up for our future

keeping our carbon expenditure below budget requires systemic changes

MRV System
(Measurement, Reporting, and Verification)

- Track emissions
- Measure progress towards carbon reduction

Carbon Tax

- Tax large carbon emitters
- Oil and gas are taxed

Renewable energy

- Scale solar-first renewables
- Electric vehicles

MRV System

(Measurement, Reporting, and Verification)

- Mandatory reporting of carbon emissions
- Progress tracking of companies' sustainability targets to ensure carbon emission is within carbon budget

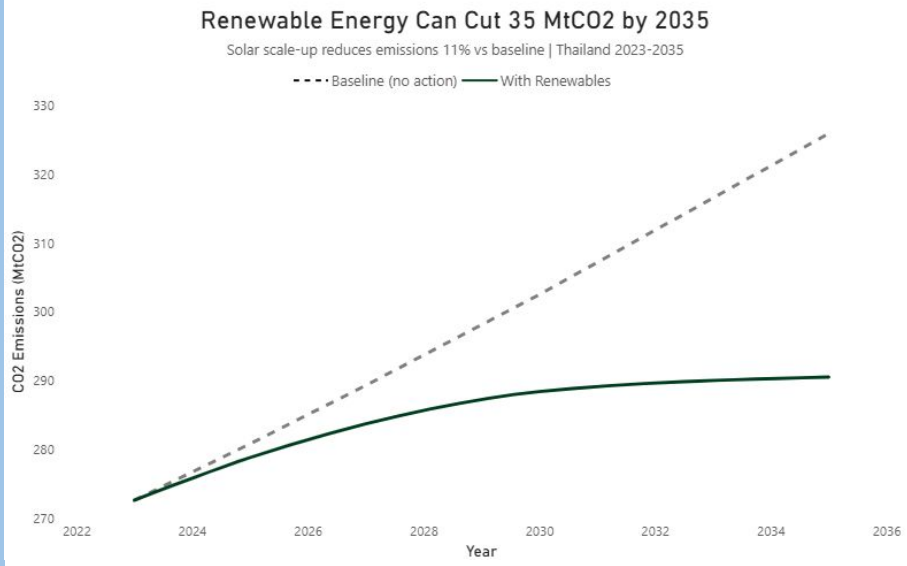
The MRV system will force companies and institutions, especially from the energy and transportation sectors to provide better reports on their carbon emissions and help enforce them to achieve their sustainability targets

Carbon Tax

- Carbon Tax: A rate of 200 THB (6.34 USD) per metric ton of CO2 now applies to upstream fossil fuels.
- Emissions Trading (ETS): Companies will be able to trade "carbon savings" as an additional revenue stream.
- Thai CBAM: Putting additional border tax on carbon-heavy imports like steel to protect the local industry.

Large carbon emitters like those from the energy and transportation sectors will be penalized because of their large use of oil and gas.

Shift to Renewable Energy



35.40

Renewables Cut by 2035

Scale Solar-first Renewables

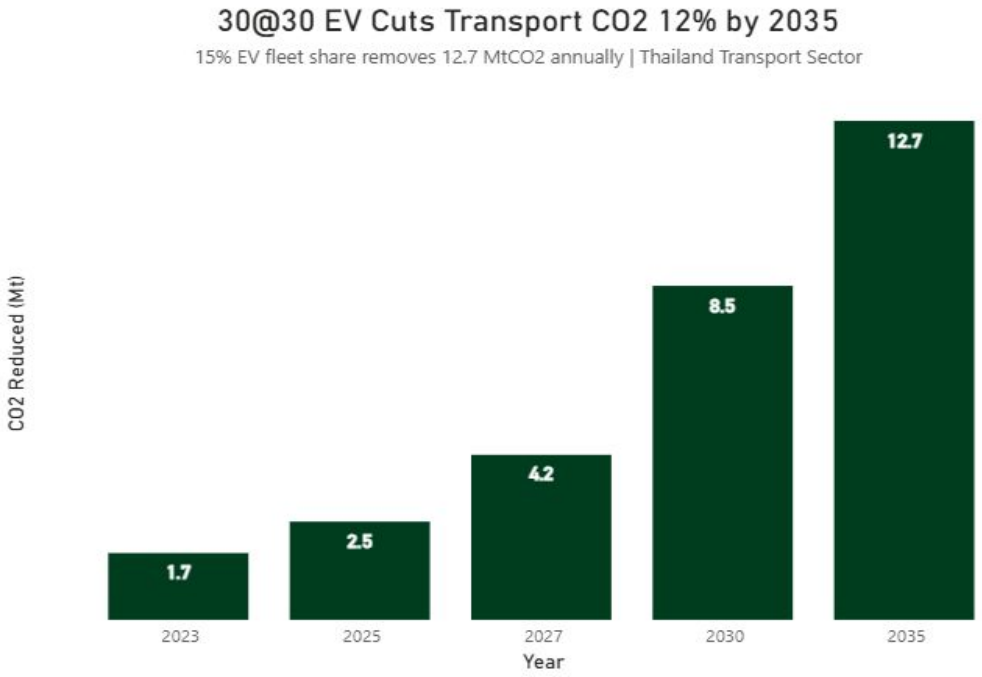
Leverage Thailand's abundant solar potential as a tropical country through utility-scale farms and industrial rooftops, supported by wind and biomass to create a diversified, low-carbon grid. Solar energy costs have plummeted and remains to be the most cost-effective energy source

Sources: OWID CO_2 Database (Thailand 2005-2023) | IEA-PVPS Thailand Solar Report 2023 | Thailand BOI 30@30 EV Policy | IPCC SR15 1.5°C Pathway | Analysis: Authors' calculations
Links: ourworldindata.org/co2 | iea-pvps.org | boi.go.th | ipcc.ch/sr15

Shift to Renewable Energy

30@30 EV

Targets 30% of total domestic vehicle production to be Zero-Emission Vehicles (ZEVs) by 2030



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the higher the damage and cost

The further we go above our carbon budget

cut emissions now and keep Thailand
on a steady path downward
by supporting Climate
Change Act